



Lab 4

Testing a Robot from the Arduino Serial Monitor

Course: RoboLink: Build and Drive Your Smartphone-Controlled Robot

Program: Mini Course Program 2026, Carleton University

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Lab Mission

In this lab, you will test a small two-wheel robot by sending commands from the Arduino IDE Serial Monitor. Your computer will send one letter at a time to the Arduino Nano. The Arduino will read the letter, control the L9110 2-channel motor driver, and move two DC motors.

Group Member Names _____

Date _____

1. Learning Goals

By the end of this lab, you should be able to:

- Send commands to an Arduino using the Serial Monitor.
- Explain why a motor driver is needed to control DC motors.
- Control two DC motors using an L9110 2-channel motor driver.
- Test robot movements: forward, backward, left, right, and stop.
- Change motor speed using PWM values from 0 to 255.
- Troubleshoot common robot problems using evidence.

2. Big Idea

The Arduino is the brain. The L9110 is the muscle.

The Arduino Nano can send control signals, but it should not power the motors directly. The L9110 motor driver safely switches motor power on and off based on the Arduino signals.



3. Materials

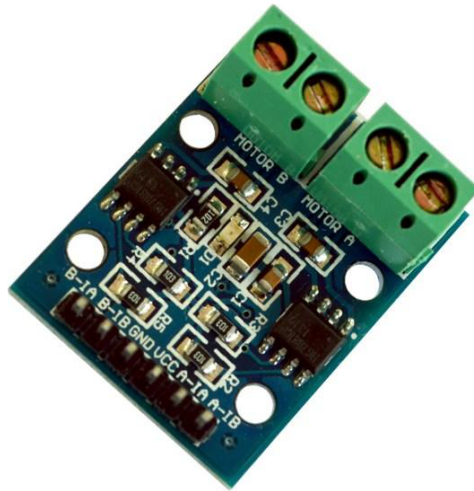
Arduino Nano	Receives commands from the computer and sends signals to the motor driver.
L9110 2-channel motor driver	Controls the direction and speed of two DC motors.
Two DC motors	Move the robot.
External motor battery or power supply	Powers the motors through the L9110 driver.
USB cable	Uploads code and opens the Serial Monitor.
Jumper wires	Connect the Arduino, driver, and power supply.

Safety and Care

Do not power the motors only from the Arduino Nano 5V pin or USB. Use an external motor battery or power supply. Always connect Arduino GND and motor battery GND together.

4. Wiring Checklist

Connect the L9110 input pins to the Arduino Nano:



L9110 2-channel motor driver. The six-pin header is labelled B-1A, B-1B, GND, VCC, A-1A, and A-1B. The green screw terminals connect to Motor A and Motor B.

L9110 Pin	Arduino Nano Pin
A-1A	D3
A-1B	D5
B-1A	D6
B-1B	D9
GND	Arduino GND and battery GND
VCC	External motor power, usually 2.5V to 12V

Before You Upload

Ask your teacher to check your wiring before turning on motor power. A wiring mistake can make the robot behave strangely or stop the upload from working.

5. Arduino IDE Setup

1. Open the Arduino IDE.
2. Open the robot code file.
3. Select **Tools** → **Board** → **Arduino AVR Boards** → **Arduino Nano**.
4. Select **Tools** → **Processor** → **ATmega168**.
5. Select the correct **Tools** → **Port**.
6. Upload the code.
7. Open **Tools** → **Serial Monitor**.
8. Set the baud rate to **9600**.

Nano Upload Tip

This lab uses an Arduino Nano with an **ATmega168** processor. If uploading fails, check that **Tools** → **Processor** → **ATmega168** is selected.

6. Serial Monitor Commands

Type one command and press Send.

Command	Robot Action
F	Move forward
B	Move backward
L	Turn left
R	Turn right
S	Stop
+	Increase speed
-	Decrease speed
?	Show the help menu

7. Experiment Procedure

Part A: First Contact

1. Open the Serial Monitor.
2. Send ?.
3. Write down what appears on the screen.

Observation: What did the Arduino print after you sent ??

Part B: Test Each Movement

Send each command and record what the robot does.

Command	Expected Movement	What Actually Happened?
F	Forward	
B	Backward	
L	Turn left	
R	Turn right	
S	Stop	

Part C: Test Speed

1. Send F.
2. Send + two times.
3. Send - two times.
4. Send S.

Observation: How did the robot change when the speed changed?

Part D: Mini Challenge

Write a sequence of commands to make your robot:

- move forward,
- turn around,
- come back,
- stop.

Your command sequence:

8. What Is Happening in the Code?

<code>setup()</code>	Runs once. It sets the motor pins as outputs, starts Serial communication, and prints the help menu.
<code>loop()</code>	Runs again and again. It checks if a new Serial Monitor command arrived.
<code>Serial.read()</code>	Reads one character from the Serial Monitor.
<code>forward()</code>	Sends PWM to the correct L9110 input pins so both motors spin forward.
<code>backward()</code>	Reverses which input pins receive PWM, so both motors spin backward.
<code>stopMotors()</code>	Sets all L9110 input pins LOW, stopping both motors.
<code>analogWrite()</code>	Sends a PWM value from 0 to 255 to control motor speed.

Why does the robot pause before moving?

The code stops the motors for 400 ms before each new movement. This gives the motor driver and motors time to settle before changing direction.

9. Expected Problems and Fixes

Problem	Possible Cause	What To Try
Nothing moves	Motor power is missing, or GND is not shared	Check L9110 VCC, battery, Arduino GND, and battery GND.
Serial Monitor shows nothing	Wrong baud rate or wrong port	Set Serial Monitor to 9600 and check Tools → Port.
Upload fails	Wrong Nano processor setting	Select Tools → Processor → ATmega168.
Robot goes backward when you send F	Motor wires are reversed	Swap the two wires on each motor.
Left and right are reversed	Motor A and Motor B are swapped, or one motor is wired backwards	Swap motor sides or reverse one motor's wires.
One motor moves but the other does not	Loose wire or wrong pin	Check D3, D5, D6, D9 and motor screw terminals.
Robot only changes direction after Stop	Motor driver needs a pause before reversing	Make sure the code includes the 400 ms pause before movement.
Robot resets when motors start	Motor current is too high for the supply	Use a stronger battery or reduce speed.

10. Reflection Questions

Answer in full sentences.

1. Why do we use an L9110 motor driver instead of connecting the motors directly to the Arduino?
2. What does the Arduino receive when you type F in the Serial Monitor?
3. What is PWM? Why is a value like 200 different from 255?

4. Why should the Arduino GND and motor battery GND be connected together?
5. If your robot moves in the wrong direction, what evidence would help you decide what to change?
6. What is one improvement you would make to this robot if you had more time?

11. Exit Ticket

In one or two sentences: What did you learn today about how computers control motion?

12. Teacher Check

Teacher Checklist

- | | |
|--|--------------|
| ☐ Wiring checked | Notes: _____ |
| ☐ Code uploaded | Notes: _____ |
| ☐ Robot responds to Serial Monitor | Notes: _____ |
| ☐ Students completed observations | Notes: _____ |
| ☐ Students answered reflection questions | Notes: _____ |